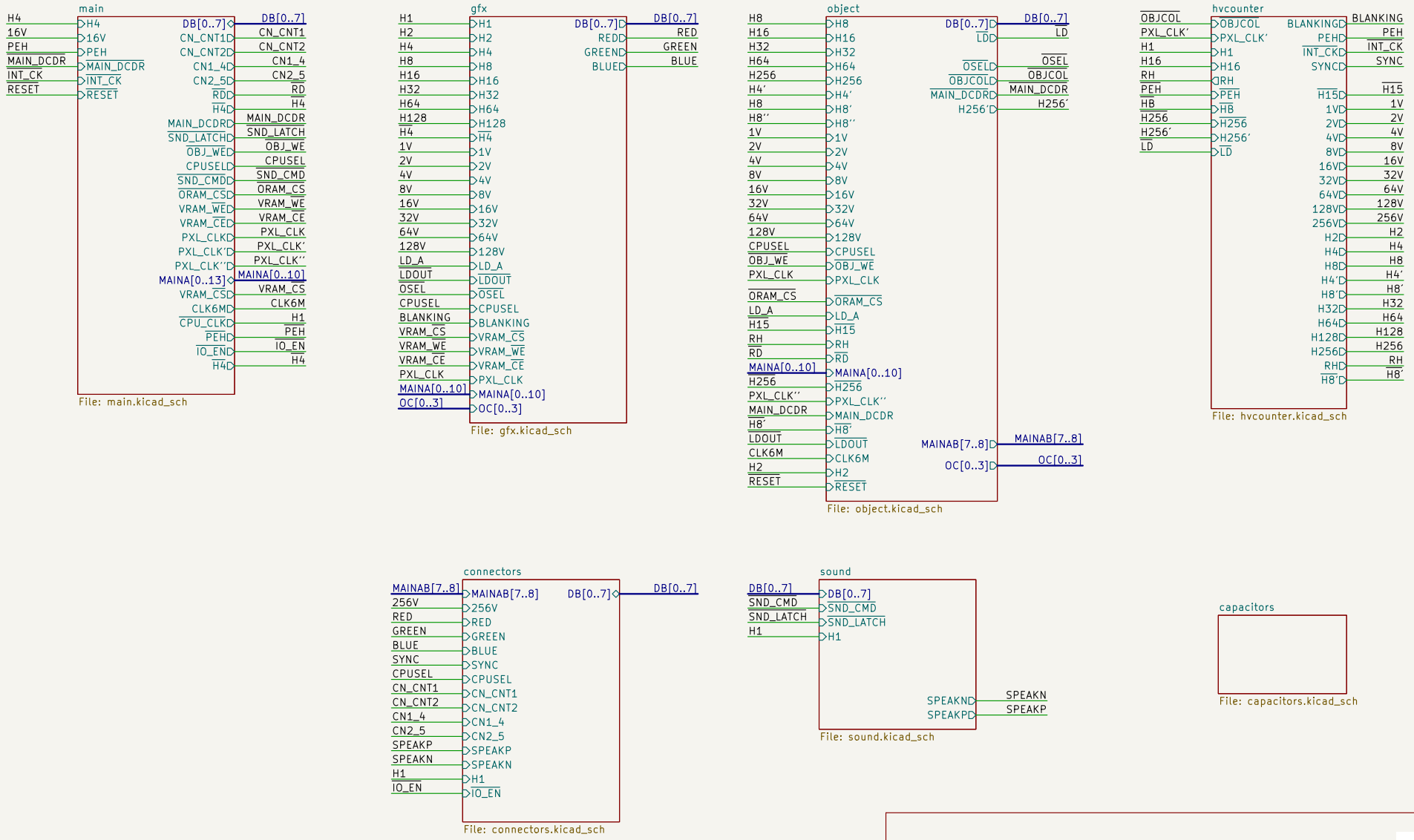
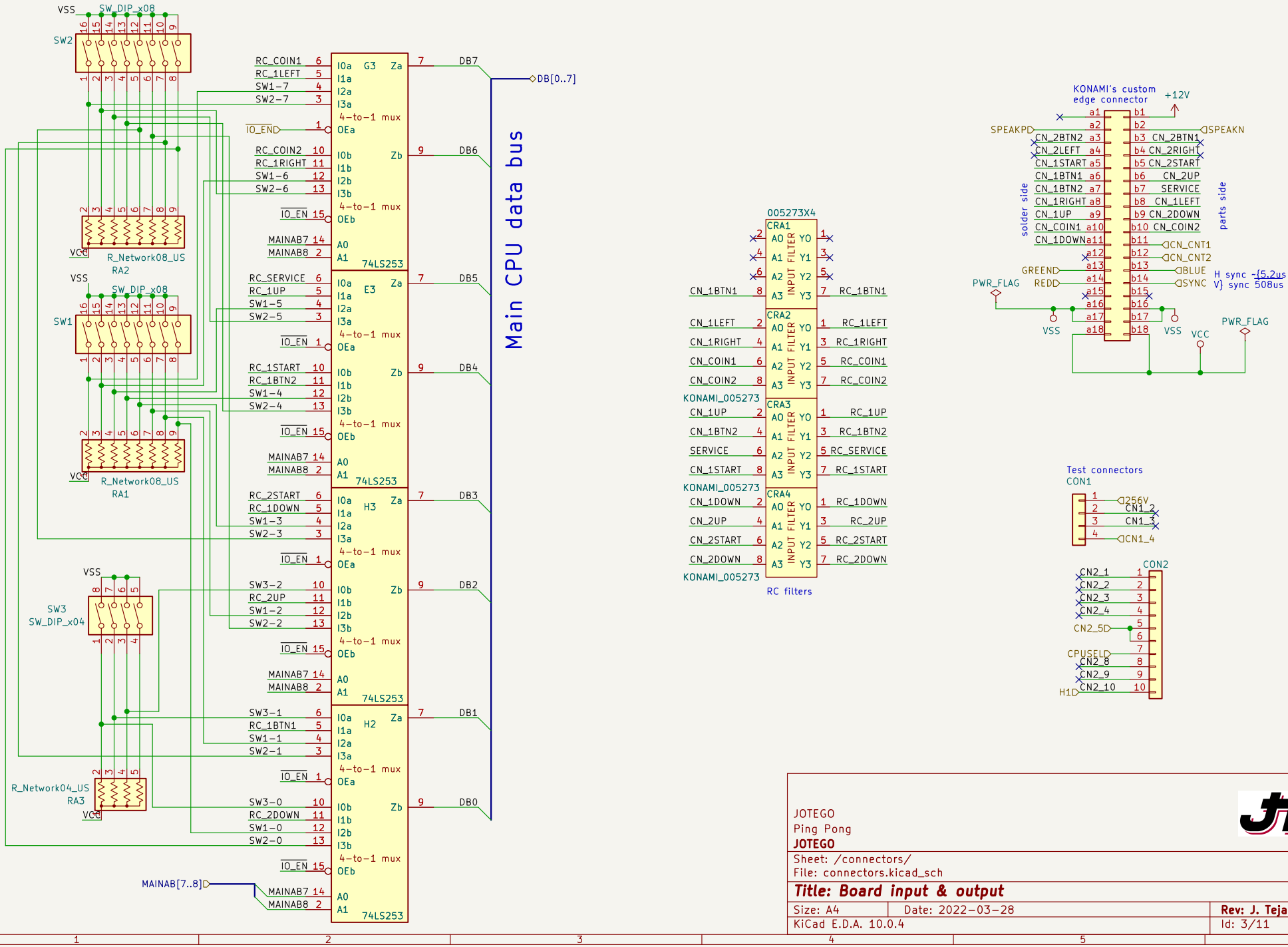

main
H4
16V
PEH
MAIN_DCDR
INT_CK
RESET
H4
16V
PEH
MAIN_DCDR
INT_CK
RESET
DB[0..7]
CN_CNT1
CN_CNT2
CN1_4
CN2_5
RD
H4
MAIN_DCDR
SND_LATCH
OBJ_WE
CPUSEL
SND_CMD
ORAM_CS
VRAM_WE
VRAM_CE
PXL_CLK
PXL_CLK'
PXL_CLK''
MAINA[0..13]
VRAM_CS
CLK6M
CPU_CLK
PEH
IO_EN
H4
DB[0..7]
CN_CNT1
CN_CNT2
CN1_4
CN2_5
RD
H4
MAIN_DCDR
SND_LATCH
OBJ_WE
CPUSEL
SND_CMD
ORAM_CS
VRAM_WE
VRAM_CE
PXL_CLK
PXL_CLK'
PXL_CLK''
MAINA[0..10]
VRAM_CS
CLK6M
H1
PEH
IO_EN
H4
File: main.kicad_sch
gfx
H1
H2
H4
H8
H16
H32
H64
H128
H4
1V
2V
4V
8V
16V
32V
64V
128V
LD_A
LDOUT
OSEL
CPUSEL
BLANKING
VRAM_CS
VRAM_WE
VRAM_CE
PXL_CLK
MAINA[0..10]
OC[0..3]
H1
H2
H4
H8
H16
H32
H64
H128
H4
1V
2V
4V
8V
16V
32V
64V
128V
LD_A
LDOUT
OSEL
CPUSEL
BLANKING
VRAM_CS
VRAM_WE
VRAM_CE
PXL_CLK
MAINA[0..10]
OC[0..3]
DB[0..7]
RED
GREEN
BLUE
DB[0..7]
RED
GREEN
BLUE
File: gfx.kicad_sch
object
H8
H16
H32
H64
H256
H4'
H8
H8''
1V
2V
4V
8V
16V
32V
64V
128V
CPUSEL
OBJ_WE
PXL_CLK
ORAM_CS
LD_A
H15
RH
RD
MAINA[0..10]
H256
PXL_CLK''
MAIN_DCDR
H8'
LDOUT
CLK6M
H2
RESET
H8
H16
H32
H64
H256
H4'
H8
H8''
1V
2V
4V
8V
16V
32V
64V
128V
CPUSEL
OBJ_WE
PXL_CLK
ORAM_CS
LD_A
H15
RH
RD
MAINA[0..10]
H256
PXL_CLK''
MAIN_DCDR
H8'
LDOUT
CLK6M
H2
RESET
DB[0..7]
LDD
OSEL
OBJCOLD
MAIN_DCDR
H256'D
DB[0..7]
LD
OSEL
OBJCOL
MAIN_DCDR
H256'
MAINAB[7..8]
OC[0..3]
MAINAB[7..8]
OC[0..3]
File: object.kicad_sch
hvcounter
OBJCOL
PXL_CLK'
H1
H16
RH
PEH
HB
H256
H256'
LD
OBJCOL
PXL_CLK'
H1
H16
RH
PEH
HB
H256
H256'
LD
BLANKING
PEH
INT_CK
SYNC
H15
1V
2V
4V
8V
16V
32V
64V
128V
256V
H2
H4
H8
H4'
H8'
H32
H64
H128
H256
RH
H8'
BLANKING
PEH
INT_CK
SYNC
H15
1V
2V
4V
8V
16V
32V
64V
128V
256V
H2
H4
H8
H4'
H8'
H32
H64
H128
H256
RH
H8'
File: hvcounter.kicad_sch
connectors
MAINAB[7..8]
256V
RED
GREEN
BLUE
SYNC
CPUSEL
CN_CNT1
CN_CNT2
CN1_4
CN2_5
SPEAKP
SPEAKN
H1
IO_EN
MAINAB[7..8]
256V
RED
GREEN
BLUE
SYNC
CPUSEL
CN_CNT1
CN_CNT2
CN1_4
CN2_5
SPEAKP
SPEAKN
H1
IO_EN
DB[0..7]
DB[0..7]
File: connectors.kicad_sch
sound
DB[0..7]
SND_CMD
SND_LATCH
H1
DB[0..7]
SND_CMD
SND_LATCH
H1
SPEAKND
SPEAKPD
SPEAKN
SPEAKP
File: sound.kicad_sch
capacitors
File: capacitors.kicad_sch
JOTEGO
Ping Pong
JOTEGO
Sheet: /
File: pingpong.kicad_sch
GX555
Title: Ping Pong
Size: A4
Date: 2022-03-28
Rev: J. Tejada
KiCad E.D.A. 10.0.4
Id: 1/11

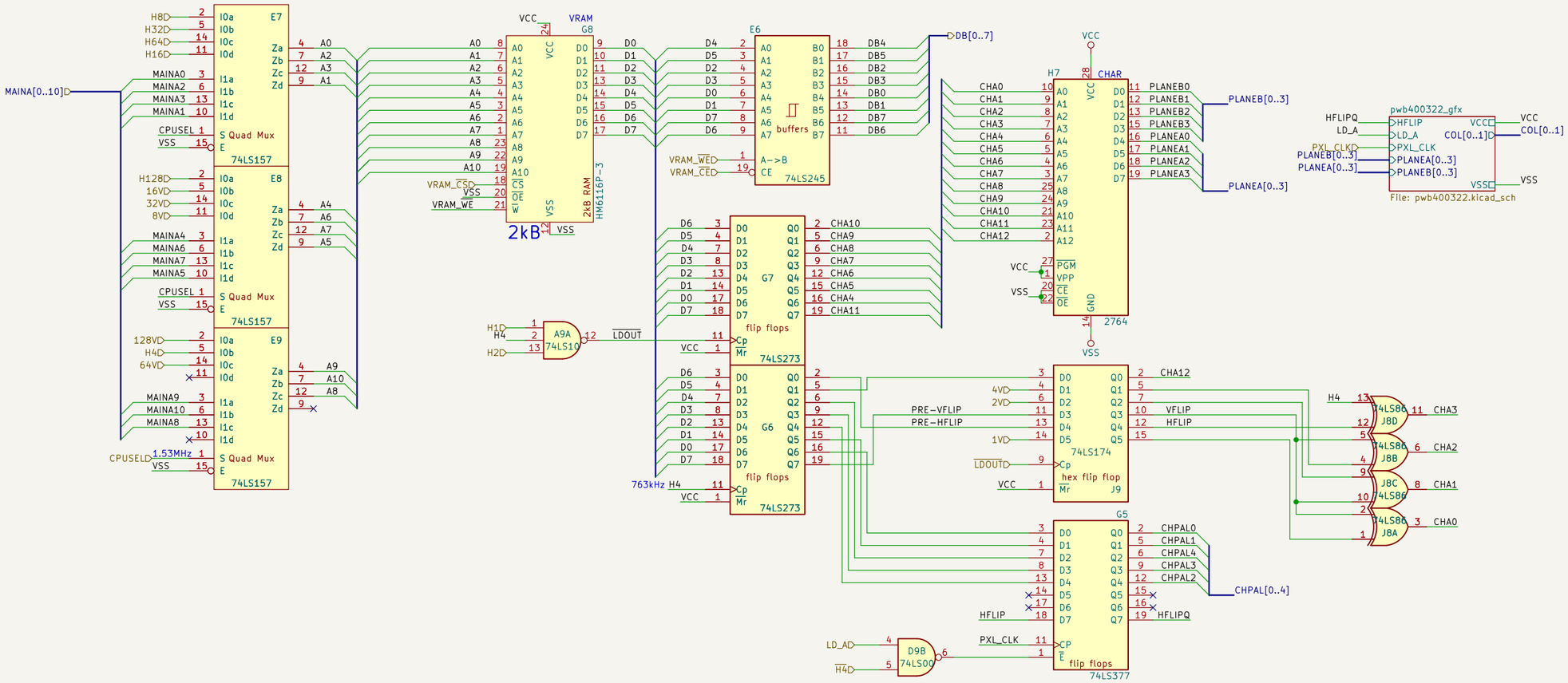
MAIN PROGRAM ROM

A5 Z80CPU
A6 27128
A7 27128
D5 KONAMI_UK
C4 74LS174
C3 74LS138
A2 74LS368
A3 74LS107
74LS139
74LS139
74LS74
74LS14
74LS138
74LS07
74LS07
74LS32
74LS00
74LS08
74LS32
74LS32
74LS07
74xx04
74LS14
74LS07
74LS174
74xx74
74LS32
74LS32
74LS00
74LS08
74LS32
74LS07
74LS07
74LS14
74LS07
74LS14
74LS07
74LS14



JOTEGO	
Ping Pong	
JOTEGO	
Sheet: /connectors/	
File: connectors.kicad_sch	
Title: Board input & output	
Size: A4	Date: 2022-03-28
KiCad E.D.A. 10.0.4	Id: 3/11

Rev: J. Tejada

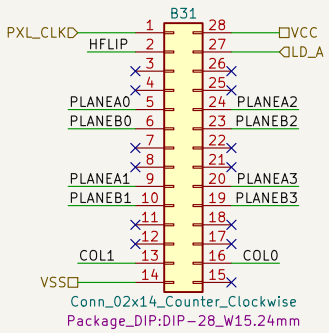


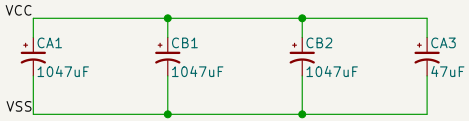
JOTEGO
Ping Pong
JOTEGO

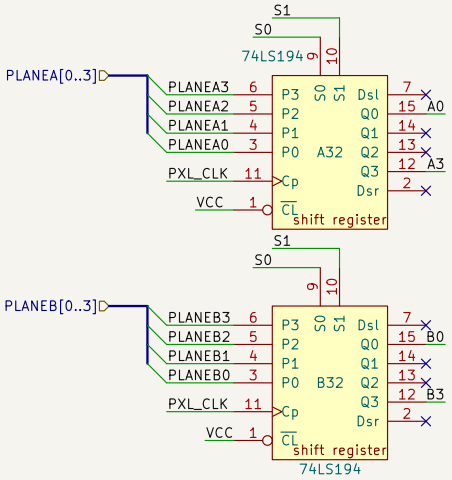
Sheet: /gfx/
File: gfx.kicad_sch

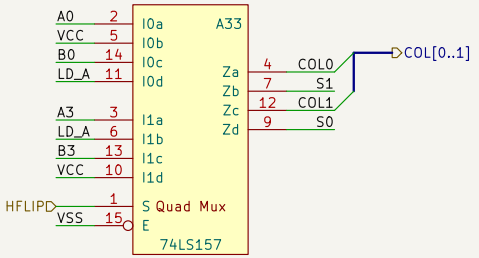
Title: Character generator

Size: A3	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 4/11









This is a small board soldered to a DIP-28 footprint. It appears twice in the design

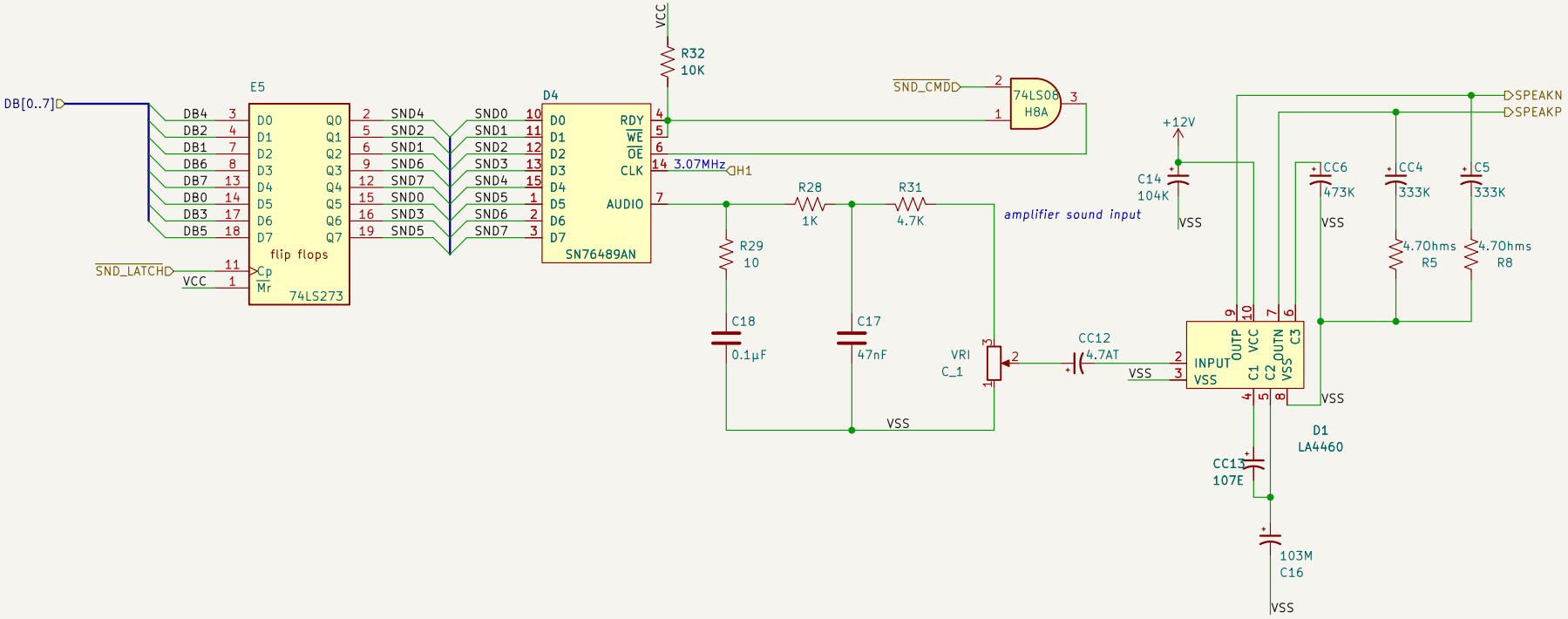
PWB-400322

JOTEGO
Ping Pong
JOTEGO

Sheet: /gfx/pwb400322_gfx/
File: pwb400322.kicad_sch

Title: Pixel shift register

Size: A4	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 5/11



JOTEGO
Ping Pong
JOTEGO

Sheet: /sound/
File: sound.kicad_sch

Title: Sound generator

Size: A4	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 6/11

OBJRAM

MAINA[0..10]

192kHz H16
781kHz H4
385kHz H8
96kHz H32

MAINA2 MAINA0 MAINA1 MAINA3 → 74LS157 (S Quad Mux, E) → AA2 AA0 AA1 AA3

16.1kHz H256
47.6kHz H64
16.1kHz RHD

MAINA6 MAINA4 MAINA5 MAINA7 → 74LS157 (S Quad Mux, E) → AA6 AA4 AA5 AA7

MAINA10 MAINA9 RDD MAINA8 → 74LS157 (S Quad Mux, E) → AA10 AA9 OEB AA8

1.54MHz CPUSEL VSS → 74LS157 (S Quad Mux, E)

VCC

A8 HM6116P-3 (2kB RAM, OBJECT TABLE)

AA0 AA1 AA2 AA3 AA4 AA5 AA6 AA7 AA8 AA9 AA10 OEB OBJ_WE VSS

DA0 DA1 DA2 DA3 DA4 DA5 DA6 DA7

24 objects read in order from the first 24x4 bytes

A10 buffers (74LS245, A->B)

DB0 DB1 DB2 DB3 DB4 DB5 DB6 DB7 → DB[0..7]

OBJ_WE1 ORAM_CS

DA1 DA3 DA5 DA7 DA6 DA4 DA2 DA0 → E11 flip flops (74LS273) → ATTR1 ATTR3 ATTR5 ATTR7 ATTR6 ATTR4 ATTR2 ATTR0

H15D VCC

PRE_LINE HFLIP ATTR4 ATTR0 ATTR1 ATTR2 ATTR3 → 74LS377 flip flops G11 → LINECS HFLIP' PAL4 PAL0 PAL1 PAL2 PAL3

PXL_CLK OBJCOL

COL[0..1]

J11 TBP24S10N Color PROM

COL0 COL1 PAL0 PAL1 PAL2 PAL3 PAL4 VSS → A0 A1 A2 A3 A4 A5 A6 A7 G2 G1 → Q0 Q1 Q2 Q3 → IC3 IC2 IC1 IC0 → IC[0..3]

E10 (74LS367)

ATTR6 ATTR7 MAINA8 MAINA7 H8' VSS → I1 I2 I3 I4 I5 I6 OEa OEb → O1a O2a O3a O4a O5b O6b → DN6 (hflip) DN7 (vflip) MAINAB8 MAINAB7 → MAINAB[7..8]

DA7 DA5 DA3 DA1 DA0 DA2 DA4 DA6 → A11 3-state flip flops (74LS374) → DN7 DN5 DN3 DN1 DN0 DN2 DN4 DN6

H8' H8''

A12 V/HFLIP XORed

DN7 DN6 DN5 DN4 DN3 DN2 DN1 DN0 → OBJV8 OBJH8 OBJH4 OBJV4 OBJV2 OBJV1 → OBJ5 OBJ4 OBJ3 OBJ2 OBJ1 OBJ0

128V 64V 32V 16V 8V 4V 2V 1V → OBJDATA OBJCOL HFLIP LINE_CS → OBJDATA OBJCOL HFLIP PRE_LINE

17383kHz H8' (input) 16766kHz H4' LD → KONAMI_503

C12 2764 8kB

OBJ0 OBJ1 OBJ2 OBJ3 OBJ4 OBJ5 OBJ6 OBJ7 OBJ8 OBJ9 OBJ10 OBJ11 OBJ12 → A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 → D0 D1 D2 D3 D4 D5 D6 D7 → OBJD0 OBJD1 OBJD2 OBJD3 OBJD4 OBJD5 OBJD6 OBJD7 → OBJD[0..7]

PGM VPP CE OE

VCC VSS OE_M

C13 flip flops (74LS273)

DA6 DA2 DA3 DA4 DA5 DA0 DA1 DA7 → D0 D1 D2 D3 D4 D5 D6 D7 → Q0 Q1 Q2 Q3 Q4 Q5 Q6 Q7 → OBJ12 OBJ8 OBJ9 OBJ10 OBJ11 OBJ6 OBJ7 OBJ13

OBJDATA

H9A 74LS74 (D, C, S, R, Q, Q̄)

ATTR5 → D

OE_OF OE_M

C11 27128 16kB ONLY FOOTPRINT

OBJ0 OBJ1 OBJ2 OBJ3 OBJ4 OBJ5 OBJ6 OBJ7 OBJ8 OBJ9 OBJ10 OBJ11 OBJ12 OBJ13 → A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 A10 A11 A12 A13 → D0 D1 D2 D3 D4 D5 D6 D7 → OBJD0 OBJD1 OBJD2 OBJD3 OBJD4 OBJD5 OBJD6 OBJD7

PGM VPP CE OE

VCC VSS OE_OF

---

pwb400322

OBJD[4..7] OBJD[0..3] PXL_CLKD HFLIP LD_AD

PLANEA[0..3] PLANEB[0..3] PXL_CLK HFLIP LD_A

VCC COL[0..1] VSS

File: pwb400322.kicad_sch

linebuffer

PXL_CLK'' DA[0..7] LINECS H2 LDD OBJCOL MAIN_DCDR H8'D LDOUT H256 CLK6M IC[0..3] RESETD

PXL_CLK'' DA[0..7] LINECS H2 LD OBJCOL MAIN_DCDR H8' LDOUT H256 CLK6M IC[0..3] RESET

OC[0..3] OSEL H256 H8'' LD_AD H256' MAIN_DCDR

OC[0..3] OSEL H256 H8'' LD_A H256' MAIN_DCDR

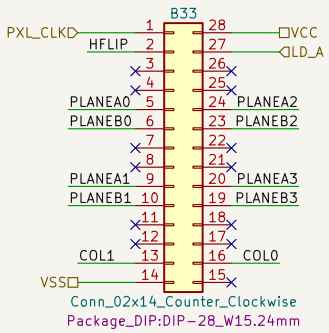
File: linebuffer.kicad_sch

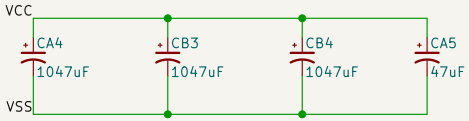
JOTEGO
Ping Pong
JOTEGO

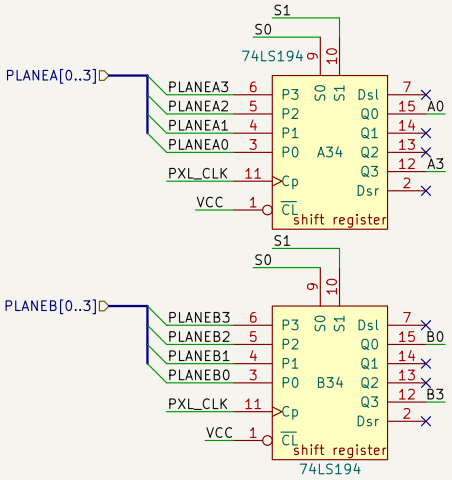
Sheet: /object/
File: object.kicad_sch

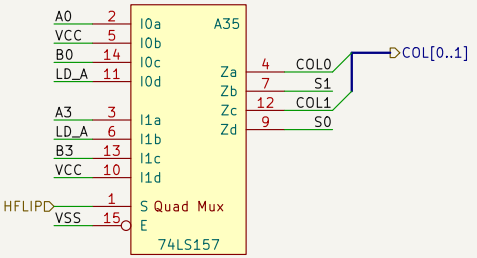
Title: Object generator

Size: A3	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 7/11









This is a small board soldered to a DIP-28 footprint. It appears twice in the design

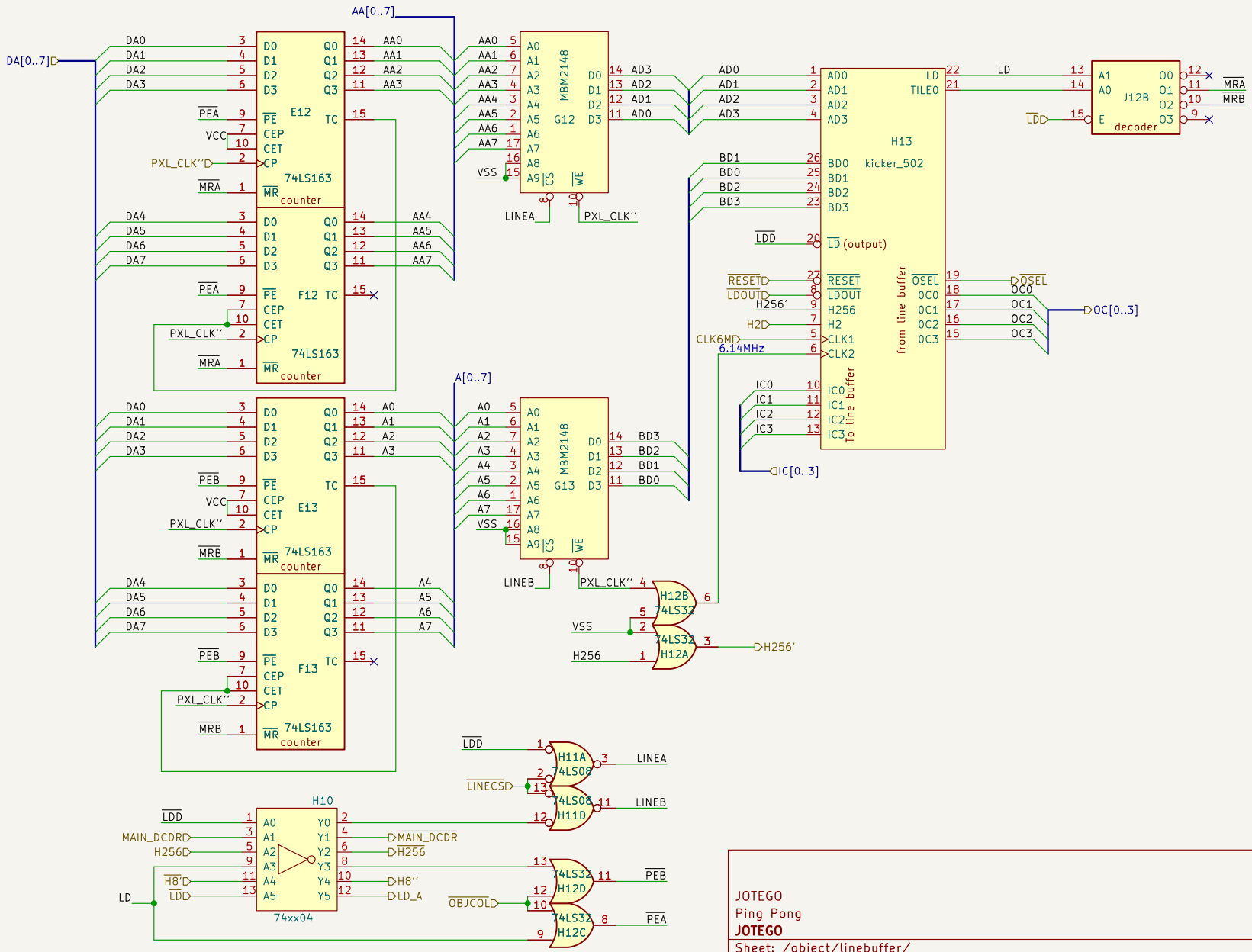
PWB-400322

JOTEGO
Ping Pong
JOTEGO

Sheet: /object/pwb400322/
File: pwb400322.kicad_sch

Title: Pixel shift register

Size: A4	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 8/11

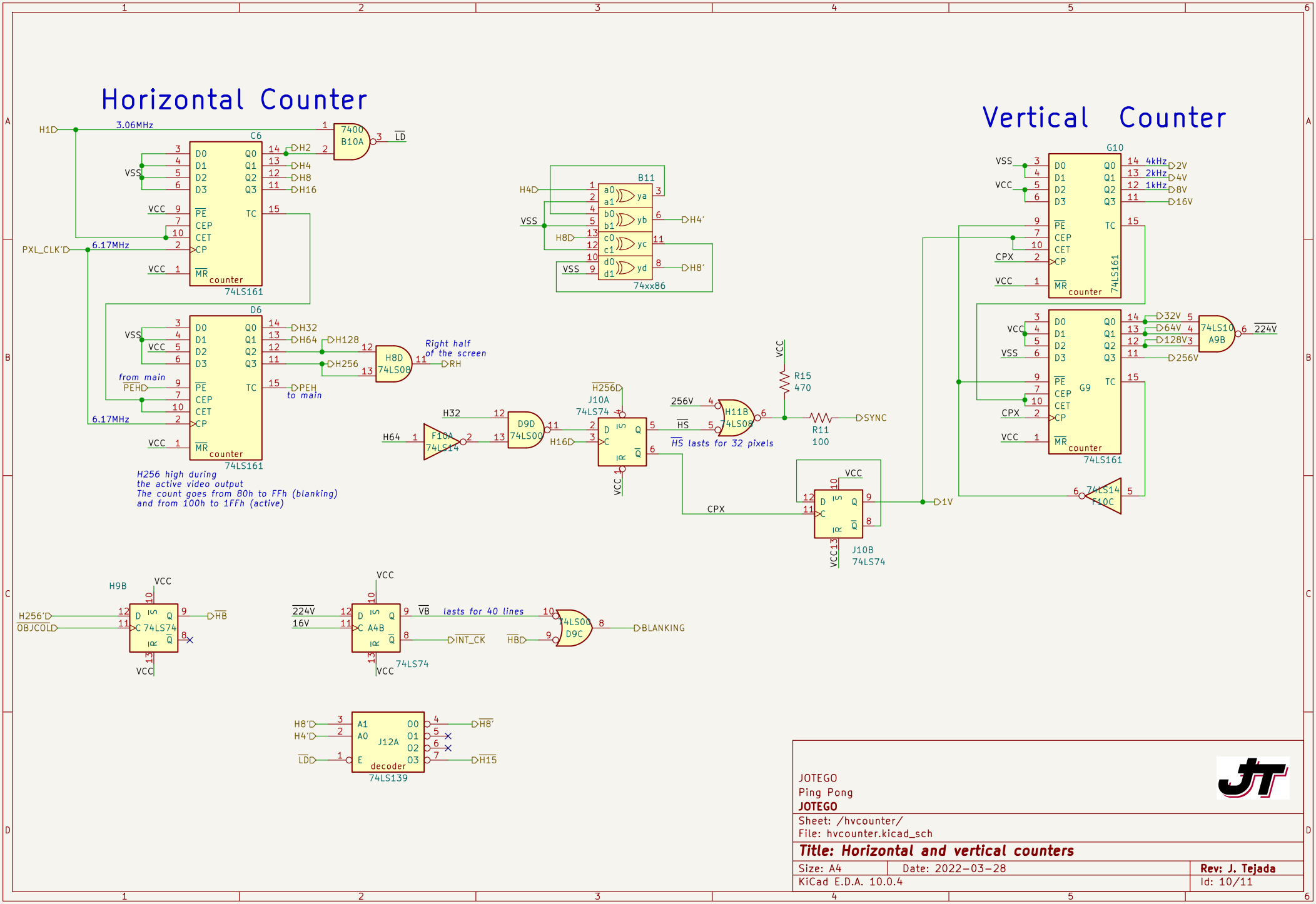


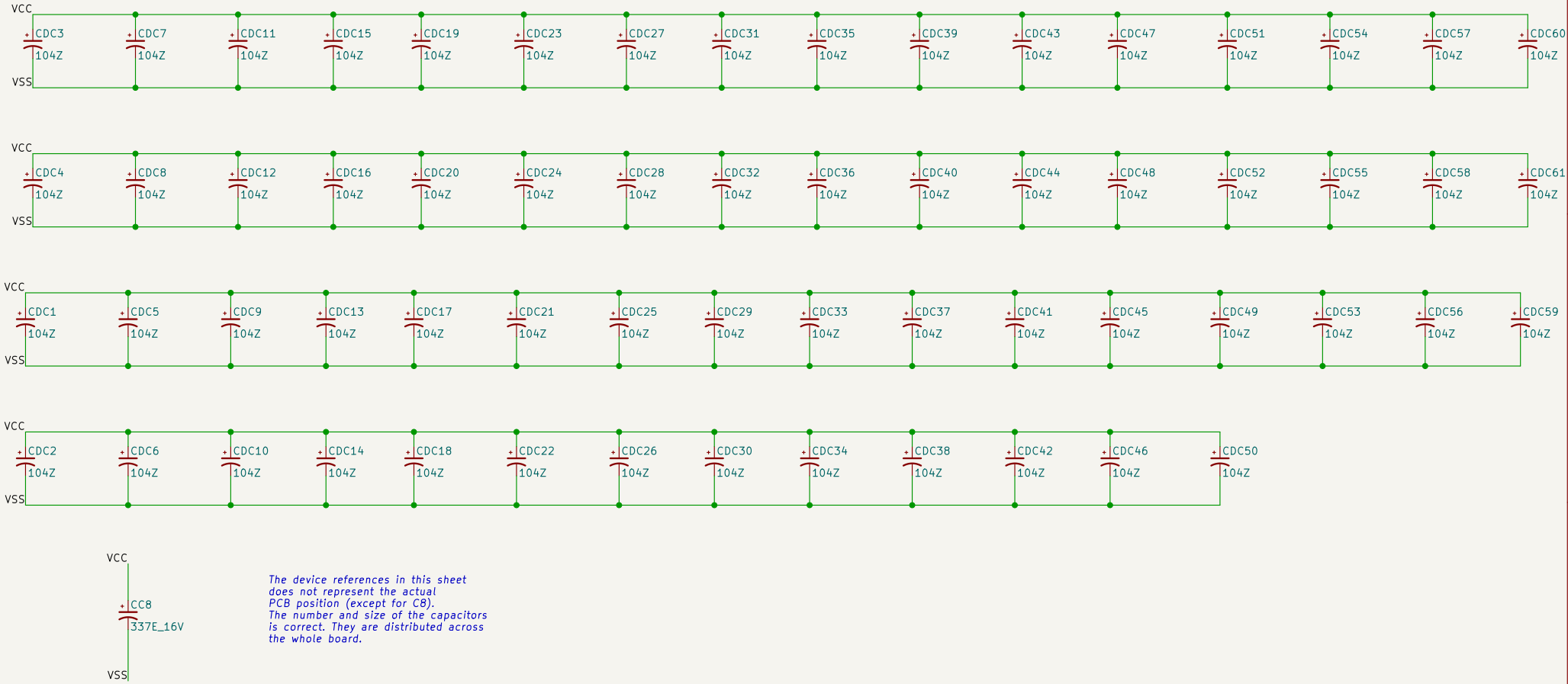
JOTEGO
Ping Pong
JOTEGO

Sheet: /object/linebuffer/
File: linebuffer.kicad_sch

Title: Doube line buffer

Size: A4	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 9/11


Horizontal Counter
Vertical Counter
H1
3.06MHz
7400
B10A
LD
C6
counter
74LS161
PXL_CLK'D
6.17MHz
VSS
VCC
H2
H4
H8
H16
D6
H32
H64
H128
H256
H8D
74LS08
Right half
of the screen
RH
from main
PEH
to main
H256 high during
the active video output
The count goes from 80h to FFh (blanking)
and from 100h to 1FFh (active)
B11
74xx86
H4'
H8'
J10A
74LS74
D9D
74LS00
F10A
74LS14
H11B
256V
HS
HS lasts for 32 pixels
R15
470
R11
100
SYNC
CPX
J10B
1V
G10
G9
2V
4V
8V
16V
4kHz
2kHz
1kHz
74LS10
A9B
224V
32V
64V
128V
256V
F10C
H9B
H256'D
OBJCOLD
HB
A4B
VB
lasts for 40 lines
INT_CK
D9C
BLANKING
J12A
decoder
74LS139
H15
JOTEGO
Ping Pong
JOTEGO
Sheet: /hvcounter/
File: hvcounter.kicad_sch
Title: Horizontal and vertical counters
Size: A4
Date: 2022-03-28
Rev: J. Tejada
KiCad E.D.A. 10.0.4
Id: 10/11



The device references in this sheet does not represent the actual PCB position (except for C8). The number and size of the capacitors is correct. They are distributed across the whole board.

JOTEGO
Ping Pong
JOTEGO

Sheet: /capacitors/
File: capacitors.kicad_sch

Title: Decoupling capacitors

Size: A4	Date: 2022-03-28	**Rev: J. Tejada**
KiCad E.D.A. 10.0.4		Id: 11/11